Ex.no:12a)

Echo client-server using tcp server.

AIM:-

To write and execute a simple java socket program the implement a TCP Echo client and echo server.

Algorithm:-

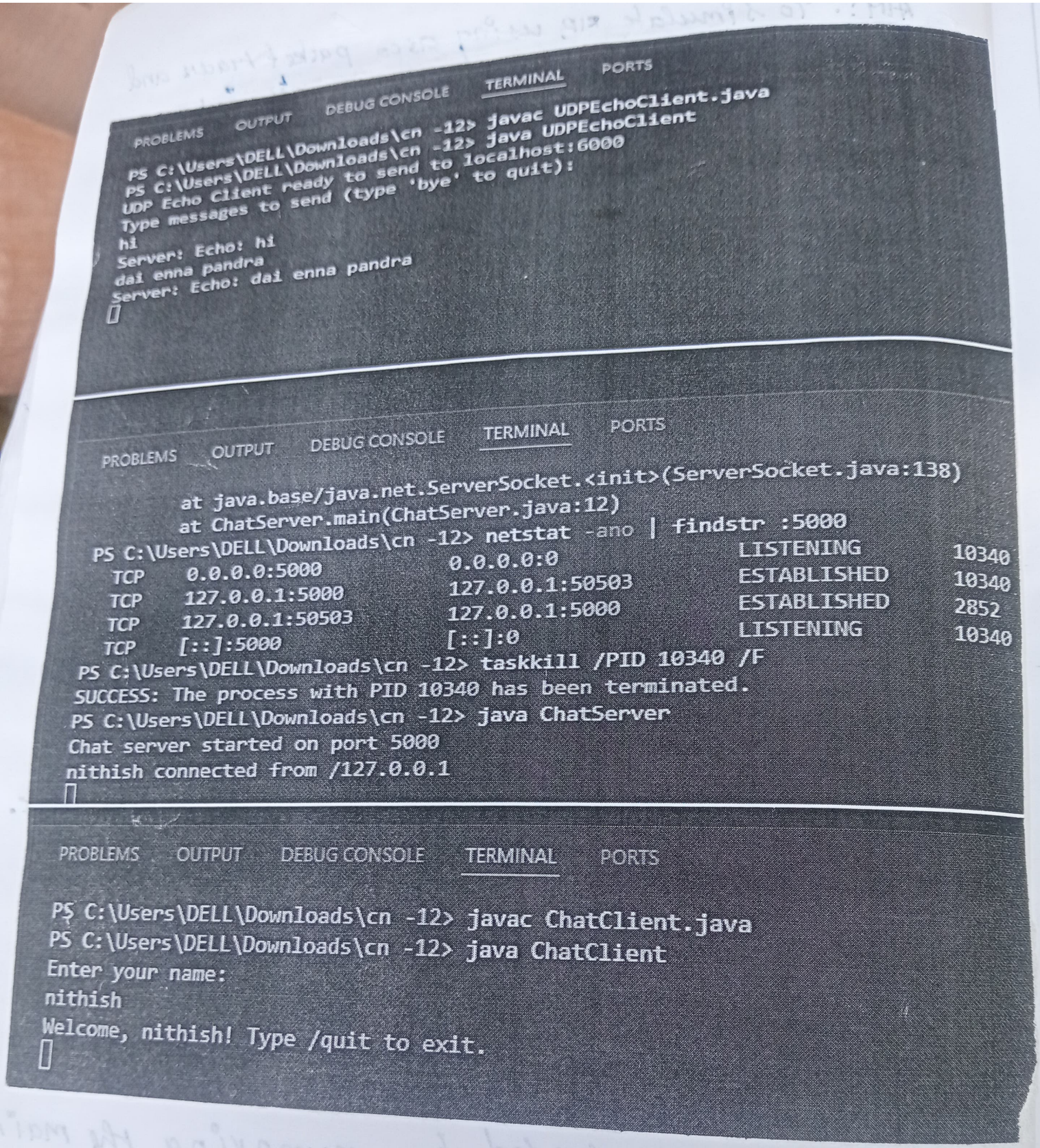
server side

1) Create a socket using socket().
2) Bind the socket the IP address and port number using bind().
3) Listen for incoming client connecting using listen().
4) Accept the Connection Request from the client using accept().
5) Read / Receive the message send by the client.
6) Send / Echo the same message back to the client
7) close the client and server sokets.

Program code:-

Echo.server.java

```
import java.io*;
import java.net.*;
public class Echo server {
    public static void main (String [] args) {
    try {
ServerSocket serverssocket = new ssocket (5000);
System.out.println ("Echo server started waiting for
                    client!");
Socket socket = serversocket.accept();
```



PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\DELL\Downloads\cn -12> javac UDPEchoClient.java
PS C:\Users\DELL\Downloads\cn -12> java UDPEchoClient
UDP Echo Client ready to send to localhost:6000
Type messages to send (type 'bye' to quit):
hi
Server: Echo: hi
dai enna pandra
Server: Echo: dai enna pandra
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
        at java.base/java.net.ServerSocket.<init>(ServerSocket.java:138)
        at ChatServer.main(ChatServer.java:12)
PS C:\Users\DELL\Downloads\cn -12> netstat -ano | findstr :5000
  TCP    0.0.0.0:5000           0.0.0.0:0              LISTENING       10340
  TCP    127.0.0.1:5000         127.0.0.1:50503        ESTABLISHED     10340
  TCP    127.0.0.1:50503        127.0.0.1:5000         ESTABLISHED     2852
  TCP    [::]:5000              [::]:0                 LISTENING       10340
PS C:\Users\DELL\Downloads\cn -12> taskkill /PID 10340 /F
SUCCESS: The process with PID 10340 has been terminated.
PS C:\Users\DELL\Downloads\cn -12> java ChatServer
Chat server started on port 5000
nithish connected from /127.0.0.1
```

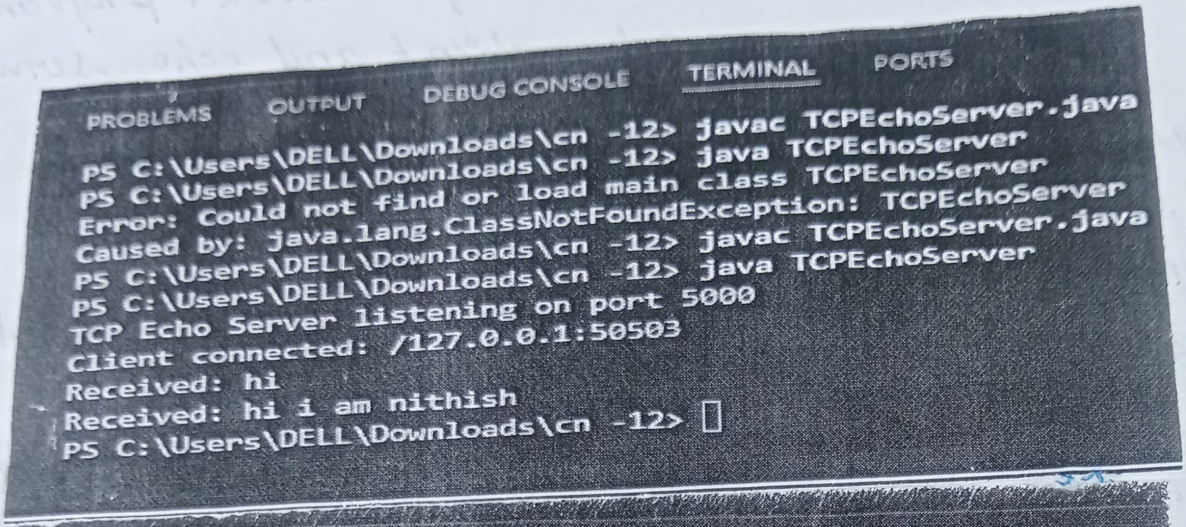
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

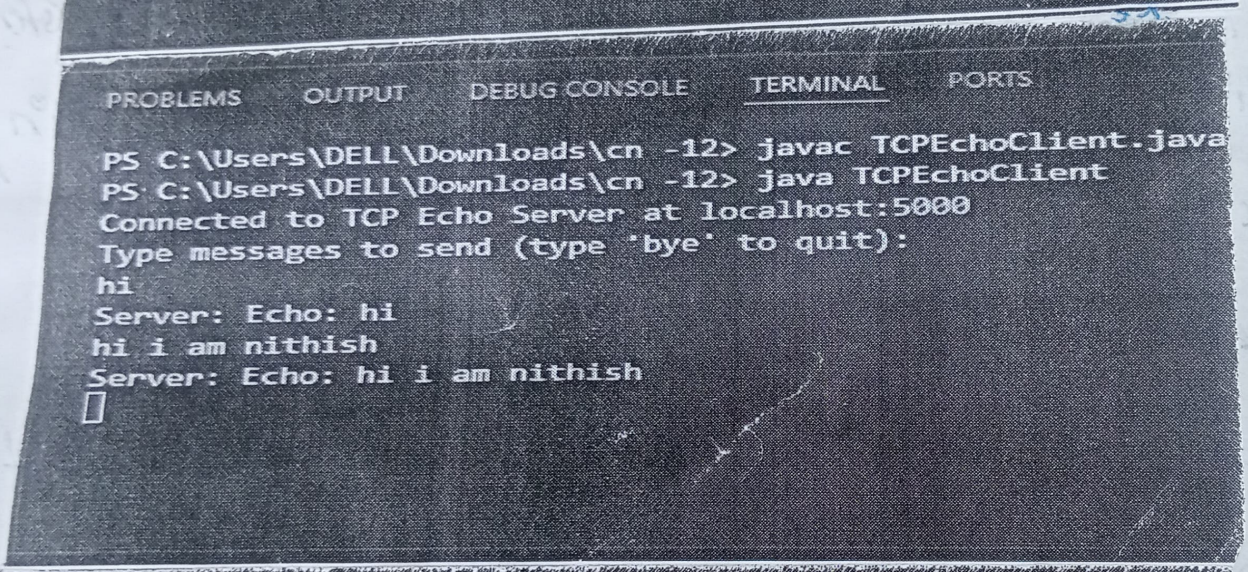
```
PS C:\Users\DELL\Downloads\cn -12> javac ChatClient.java
PS C:\Users\DELL\Downloads\cn -12> java ChatClient
Enter your name:
nithish
Welcome, nithish! Type /quit to exit.
```

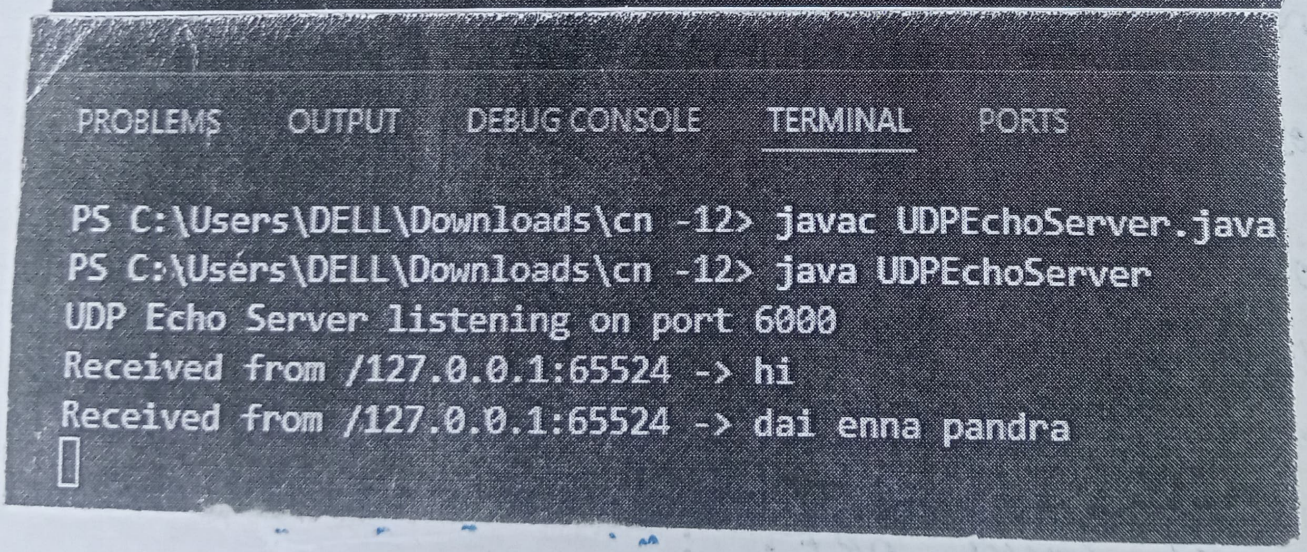
```
        System.out.println("Client Connected!");
Buffer Reader in = New Buffer Reader(new Input
                Stream Reader(Socket.get Input Stream()));
print writer out = new print writer(Socket.get
                output Stream(), true);
String inputLine = in.readLine();
System.out.println = in.readline();
Socket.close()
Server Socket.close();
} catch (Io Exception e) {
                e.print Stack Trace();
    }
}
}
```

Echo client.java

```
import java.io.*;
import java.net.*;
public class Echo client {
    public static void main(String[] args){
    try {
    Socket socket = new Socket("localhost", 5000);
System.out.println("Connect to Echo server!");
Buffer Reader user Input = new Buffer Reader(New
                Input Stream Reader(System.
                in));
```


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\DELL\Downloads\cn -12> javac TCPEchoServer.java
PS C:\Users\DELL\Downloads\cn -12> java TCPEchoServer
Error: Could not find or load main class TCPEchoServer
Caused by: java.lang.ClassNotFoundException: TCPEchoServer
PS C:\Users\DELL\Downloads\cn -12> javac TCPEchoServer.java
PS C:\Users\DELL\Downloads\cn -12> java TCPEchoServer
TCP Echo Server listening on port 5000
Client connected: /127.0.0.1:50503
Received: hi
Received: hi i am nithish
PS C:\Users\DELL\Downloads\cn -12>


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\DELL\Downloads\cn -12> javac TCPEchoClient.java
PS C:\Users\DELL\Downloads\cn -12> java TCPEchoClient
Connected to TCP Echo Server at localhost:5000
Type messages to send (type 'bye' to quit):
hi
Server: Echo: hi
hi i am nithish
Server: Echo: hi i am nithish


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\DELL\Downloads\cn -12> javac UDPEchoServer.java
PS C:\Users\DELL\Downloads\cn -12> java UDPEchoServer
UDP Echo Server listening on port 6000
Received from /127.0.0.1:65524 -> hi
Received from /127.0.0.1:65524 -> dai enna pandra

```
BufferReader in = new BufferReader(new InputStreamReader
                  (socket.getInputStream()));
System.out.print("Enter message to Echo");
String message = userInput.readLine();
out.println(message);
String echoResponse = in.readLine();
System.out.println("Server Echo Response:" + echoResponse);
socket.close();
} catch (IOException e) {
        e.printStackTrace();
    }
  }
}
```

Out:

Result:

The TCP Echo Client/Server program was successfully implemented, compiled, and executed and the server correctly echoed the received message back to the client.